Nutritional Assessment Tools for Diabetes Management

CBME Practical Guide for MBBS Community Medicine Posting

NUTRITIONAL ASSESSMENT FRAMEWORK

Why Nutritional Assessment Matters

- **Diabetes Management**: Nutrition is the cornerstone of diabetes care
- **Cultural Context**: Indian diet patterns affect glycemic control
- **Economic Factors**: Cost of healthy foods impacts adherence
- **Community Level**: Household food security affects diabetes outcomes

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ASSESSMENT METHODS

1. 24-Hour Diet Recall

Purpose: Assess actual dietary intake and patterns **How to Conduct**: Ask patient to recall all foods eaten in past 24 hours

Steps:

1. **Create comfortable environment**: Private, non-judgmental setting
2. **Use memory aids**: Calendar, clock for timing
3. **Quantify portions**: Use common household measures
4. **Probe for details**: Cooking methods, added oils/sugars
5. **Validate recall**: Compare with typical day's intake

Questions to Ask:

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- What time did you wake up yesterday?
- What was your first meal/drink of the day?
- Where did you eat each meal?
- Who prepared the food?
- Any snacks between meals?
- Were there any special occasions?
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2. Food Frequency Questionnaire (FFQ)

Purpose: Assess usual dietary patterns and habits over time

Indian-Focused Categories:

Carbohydrates/Staples

- Rice consumption: □ Daily □ Weekly □ Monthly
- Chapati/Roti: How many per meal?
- Bread/Biscuits: □ White □ Brown □ Never

Proteins

- Eggs: □ Daily □ Weekly □ Monthly □ Never
- Chicken/Fish/Mutton: Frequency per week?
- Dairy: Milk/Milk products daily quantity?
- Lentils/Pulses: Types and frequency?

Fruits and Vegetables

- Raw vegetables daily serving?
- Cooked vegetables type and quantity?
- Fruits daily serving (seasonal vs year round)?

Fats and Sweets

- Ghee/Oil type and quantity used daily?
- Sweets/Soft drinks frequency?
- Fried foods consumption per week?

Beverages

- Tea/Coffee daily cups with milk/sugar?
- Sugared drinks frequency?
- Traditional drinks (buttermilk, etc.)?

DIETARY ANALYSIS TOOLS

Carbohydrate Counting

Indian Context Guidelines:

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Rice/Roti: 1.5 katori = 45-60g carbs = 3-4 carb exchanges
Legumes: 1 katori cooked = 15-20g carbs = 1 carb exchange
Milk: 1 cup (240ml) = 12g carbs = 1 carb exchange
Fruit: 1 medium apple/guava = 15-20g carbs = 1 carb exchange
Vegetables: Unlimited except potato, beet, carrot
```

Exchange System:

- **Carb Exchanges**: 1 exchange = 15g carbs = 1 unit insulin typically
- **Protein Exchanges**: 1 exchange = 7g protein

- **Fat Exchanges**: 1 exchange = 5g fat

Glycemic Load Assessment

High GI Foods (>70): White rice, white bread, potatoes, cornflakes **Medium GI Foods (56-60)**: Whole wheat chapati, brown rice, sweet potato **Low GI Foods (<55)**: Most non-starchy vegetables, pulses, most fruits

Indian Diet Glycemic Considerations:

- Rice variety affects GI (Basmati rice higher than brown rice)
- Cooking method matters (soaked rice lower GI than boiled)
- Food combinations change GI (vegetables with rice lower overall GL)

NUTRITION CARE PROCESS

Four Steps Process

Step 1: Nutrition Assessment

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Anthropometric: BMI, waist circumference, weight trends
Biochemical: HbA1c, lipid profile, liver/kidney function
Clinical: Dental health, GI symptoms, activity level
Dietary: 24hr recall, FFQ, food security assessment
Environmental: Kitchen facilities, family cooking roles
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Step 2: Nutrition Diagnosis

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- Excessive carbohydrate intake (identified from FFQ)
- Inadequate dietary intake for weight management
- Food insecurity affecting medication adherence
- Limited knowledge of diabetes nutrition principles
- Food medication interactions
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Step 3: Nutrition Intervention

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- Nutrition Prescription: 1500 kcal, 45-60% carbs, <7% saturated fat
- Education: Carbohydrate counting, label reading
- Coordination: Family cooking changes, physician coordination
- Environment: Social support systems, realistic goals
```

Step 4: Nutrition Monitoring & Evaluation

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- Outcome measures: Weight change, HbA1c trends
- Process measures: SMBG patterns, meal timing adherence
- Factors affecting outcome: Stress, illness, medication changes
- Reassessment time frame: Monthly for first 3 months, then quarterly
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CULTURALLY APPROPRIATE COUNSELING

Indian Dietary Patterns

North Indian Diet Pattern

- Staple: Wheat (chappati, paratha)
- Strengths: Dairy-rich, vegetable variety
- Challenges: Ghee generosity, sweets culture
- Opportunity: Substitute wheat for bajra/ragi

South Indian Diet Pattern

- Staple: Rice (multiple varieties)
- Strengths: Lentil-rich, coconut disadvantages
- Challenges: Rice dominance, low fiber intake
- Opportunity: Brown rice, millets, vegetable increase

Festive and Social Eating

- Festivals: High sugar/high fat foods common
- Social functions: Buffet-style portion distortion
- Religious practices: Fasting days vs. festival days
- Solution: Pre-planning, moderation strategies, cultural alternatives

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PRACTICAL TOOLS FOR STUDENTS

Portion Size Estimation

Household Measures:

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Rice/Chappati: Fist-size portion
Protein: Palm-size portion
Vegetables: Two fists
Fruits: One fist
Oil: Thumb-size portion
```

Visual Cues:

- Chapati: CD-sized circle = 120 kcal

- Rice: Cricket ball = 200 kcal
- Vegetable curry: Fist = 50 kcal
- Fruit: Tennis ball = 60 kcal

Menu Planning Tool

Individualized Meal Plan Framework:

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Breakfast (300-400 kcal): 2 carb + 1 protein + ½ fruit exchanges
Mid-morning Snack (150-200 kcal): 1 fruit exchange
Lunch (500-600 kcal): 4 carb + 2 protein + 2 vegetable exchanges
Evening Snack (150-200 kcal): 1 carb + 1 protein exchange
Dinner (400-500 kcal): 3 carb + 2 protein + vegetable exchanges
```

Blood Glucose Response Prediction

Meal Glycemic Impact:

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Low Impact (<70 mg/dL rise): Vegetable + protein combination
Moderate Impact (70-130 mg/dL rise): Mixed meal with carbs
High Impact (>130 mg/dL rise): Simple carbs alone, sugary foods
```

PERFORMANCE MEASURES

Nutrition Counseling Skills Checklist

For OSCE/Direct Observation Assessment:

1. **Assessment Skills** □ Nutrition-focused history taking □ Anthropometric measurements accurate □ 24-hour recall completed thoroughly □ Cultural/dietary preferences explored

2. **Counseling Communication** □ Diet recommendations individualized □ Cultural respect demonstrated □ Family involvement discussed □ Realistic goals set collaboratively

3. **Education Effectiveness** □ Carbo-hydrate counting taught □ Portion sizes demonstrated □ Label reading skills explained □ Problem-solving barriers addressed

4. **Follow-up Planning** □ Nutrition goals documented □ Timeline for reassessment set □ Support systems identified □ Resources for continued learning provided

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QUALITY IMPROVEMENT TOOLS

Monthly Camp Nutrition Assessment

Camp-Level Data Collection:

- Average meals per day requiring insulin adjustment
- Food insecurity prevalence among screened patients
- Common dietary myths identified
- Referral to nutritionist completion rate

Action Items Based on Data:

- New education materials for common myths
- Partnership with local nutritionists
- Food security screening integration
- Community cooking demonstration camps

Student Performance Tracking

Self-Assessment Questions:

- Can I explain carbohydrate counting to a patient?
- Do I recognize portion sizes accurately?
- Can I adapt dietary advice to local food availability?
- Do I assess family cooking patterns appropriately?

RESOURCES FOR MEDICAL STUDENTS

Reference Materials

- **Exchange Lists for Indian Foods** by NIN (National Institute of Nutrition)
- **Meal Planning for Diabetes** - Simplified Indian version
- **Cultural Adaptation Guidelines** for diabetes nutrition education
- **Food Composition Tables** for South Asian foods

Online Resources

- **IDF South Asia**: www.idf.org/our-network/regions-members/south-east-asie
- **National Institute of Nutrition**: www.nin.res.in
- **Indian Council of Medical Research**: diabetes management guidelines
- **RSSDI Educational Resources**

IMPLEMENTATION NOTES

Training Requirements

- **Basic Training**: Diabetes nutrition essentials (2 hour session)
- **Advanced Training**: Complex cases and cultural adaptation (4 hours)
- **Practice Sessions**: Role-playing nutritional counseling (2 hours)
- **Skill Assessment**: Direct observation during community camps

Cost Considerations

- **Free Resources**: NIN booklet, ICMR guidelines

- **Low-Cost Tools**: Household measures demonstration cards
- **Sustainable Approach**: Train local health workers for ongoing support

Integration with Healthcare System

- **PHC Level**: Basic nutrition assessment and counseling
- **Referral Level**: Dietician for complex cases, food allergies
- **Community Level**: Cooking demonstrations, support groups
- **Quality Assurance**: Regular audit of nutritional counseling effectiveness

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Nutrition Assessment Guide: This practical tool helps MBBS students assess nutritional status, provide dietary counseling, and contribute to comprehensive diabetes care in community settings.

Faculty Facilitation: Use this guide during small group sessions and community camps to build nutritional assessment competencies essential for diabetes management.